

Advanced Data Analysis



These are the same subjects measured before (pre_score) and after (post_score) an intervention. Did their scores change significantly?

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Because these are the **same subjects measured twice** (before and after the intervention), the appropriate analysis is a **paired-samples test**.

First, I checked whether the difference scores were approximately normally distributed. The normality assumption was violated (Shapiro–Wilk $p < 0.001$), so a **Wilcoxon signed-rank test** is the more appropriate choice than a paired t-test.

Results

- Mean pre-score = **48.49**
- Mean post-score = **52.82**
- Mean change = **+4.33 points**

Wilcoxon Signed-Rank Test

The intervention produced a statistically significant change in scores:

- Test statistic $W = 19.0$
- p-value = **0.000018**

Since $p < 0.05$, we conclude that the scores **changed significantly after the intervention**.

The post-intervention scores were generally **higher** than the pre-intervention scores, suggesting the intervention improved scores.



